**Subject Description Form**

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| **Subject Code** | ITC6863RT |
| **Subject Title** | Guided Study in Smart Textiles & Clothing |
| **Credit Value** | 3 |
| **Level** | 6 |
| **Pre-requisite / Co-requisite / Exclusion** | Nil |
| **Objectives** | This subject will provide a guide for research students to carry out in-depth study in smart technologies for textiles and clothing. |
| **Intended Learning Outcomes** | Upon completion of the subject, students will be able to:<br><br>a. Chart the development roamap of the field<br>b. Comprehend scientific principles of the smart textiles<br>c. Use analytic method and characterization techniques for smart textiles<br>d. Envisage applications of smart textiles and clothing |
| **Subject Synopsis/ Indicative Syllabus** | This subject will provide a guide for research students to carry out in-depth study in smart technology for textiles and clothing, including principles, sensors and actuators, signal transmission and processing, multi-functional design, integration processes as well as analysis and evaluation. |
| **Teaching/Learning Methodology** | Class studies, laboratories and self-studies will be conducted on the field and specific topics, e.g. thermal sensitive textiles and clothing, conductive fibers, shape memory alloy and polymers, photonic fibers and fabrics, wearable electronics and photonics. Group discussions and presentation will be conducted. |
| **Assessment Methods in Alignment with Intended Learning Outcomes** | (see table below) |

| Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) | | | | | |
|---|---|---|---|---|---|---|---|
| | | a | b | c | d | | |
| 1. Coursework | 40% | ✓ | ✓ | | | | |
| 2. Examination | 60% | | ✓ | ✓ | ✓ | | |
| Total | 100% | | | | | | |

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Coursework is used to train and assess students' knowledge and skills learned in this subject. Examination is designed to assess the overall learning outcomes.

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| **Student Study Effort Expected** | Class contact: | |
| | ▪ Tutorials | 5 Hrs |

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|---|---|---|
| | ▪ Presentation/discussions | 16 Hrs |
| | ▪ Laboratory demonstration | 5 Hrs |
| | Other student study effort: | |
| | ▪ Literature studies | 44 Hrs |
| | ▪ Projects | 50 Hrs |
| | Total student study effort | 120 Hrs |
| **Reading List and References** | <u>Monographs</u>:<br>1. Tao, X.M., 2001. *Smart fibres, fabrics and clothing.* Cambridge: Woodhead Publishing Co.<br>2. Tao X.M., 2005. *Wearable electronics and photonics.* Cambridge: Woodhead Publishing Co.<br>3. Vladan K., 2016. *Smart textiles and their applications*. Duxford: Woodhead Publishing Co.<br>4. Dominique P., Pierre C., 2019. *Wearables, smart textiles & smart apparel*. ISTE Press Ltd.<br>5. Stefan S., Oliver A., 2017. *Smart textiles: fundamentals, design, and interaction*. Cham: Springer.<br>6. Tilak D., 2015. *Electronic textiles: smart fabrics and wearable technology*. Cambridge: Woodhead Publishing Co.<br>7. Mattila H.R., 2006. *Intelligent Textiles and Clothing*, Cambridge: Woodhead.<br>8. Seymour, S., 2010. *Functional aesthetics: visions in fashionable technology*. New York. Springer.<br>9. Hu, J.L., 2011. *Adaptive and Functional Polymers, Textiles and Their Applications*, London: Imperial College Press.<br><br><u>Journals</u>:<br>1. Science<br>2. Nature<br>3. Natural Materials<br>4. Advanced Materials<br>5. Advanced Functional Materials<br>6. ACS Nano<br>7. Chemistry of Materials<br>8. Applied Physics Letters<br>9. Textile Research Journal | |